REAL SOURCE DOCUMENT — MIT COURSE CATALOG

Institution: Massachusetts Institute of Technology (MIT)

Course: 6.1220[J] — Design and Analysis of Algorithms (formerly 6.046[J], cross-listed as 18.410[J])

Source URL: https://catalog.mit.edu/subjects/6/ (MIT Course 6 subject listings)

Supporting source: https://catalog.mit.edu/search/?P=6.1220

Date Accessed: March 29, 2025

Department: Electrical Engineering and Computer Science (EECS) / Mathematics (Course 18)

COURSE DESCRIPTION: 6.1220[J] Design and Analysis of Algorithms

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OFFICIAL CATALOG ENTRY:

Same subject as 18.400[J]

Prereq: (6.1200[J] and 6.1210) or permission of instructor

Level: U (Undergraduate)

Offered: Spring

Units: 4-0-8 (12 units)

Instructors: E. Demaine, M. Goemans, S. Raghuraman (representative; may vary)

DESCRIPTION:

Techniques for the design and analysis of efficient algorithms, emphasizing methods useful in practice. Topics include sorting; search trees, heaps, and hashing; divide-and-conquer; dynamic programming; greedy algorithms; amortized analysis; graph algorithms; and shortest paths. Advanced topics may include network flow; computational geometry; number-theoretic algorithms; polynomial and matrix calculations; caching; and parallel computing.

PREREQUISITE ANALYSIS:

- Required: 6.1200[J] (Mathematics for Computer Science) — must be completed
- Required: 6.1210 (Introduction to Algorithms) — must be completed
- Alternative: "or permission of instructor" — instructor consent can substitute for either prerequisite
- This is an AND requirement: both 6.1200[J] AND 6.1210 are required (not either/or)

NOTE: This course is listed as "Same subject as 18.400[J]" — students may register under either number and receive equivalent credit. Students in Course 18 (Mathematics) typically register as 18.400[J].

PREREQUISITE CHAIN FROM SCRATCH:

To reach 6.1220[J] from no prerequisites:

Step 1: 6.100A (Introduction to CS Programming) — no prereq

Step 2 (concurrent or sequential): 6.1200[J] (Mathematics for CS) — prereq: Calculus I GIR (18.01)

Step 3: 6.1210 (Introduction to Algorithms) — prereq: 6.100A + 6.1200[J]

Step 4: 6.1220[J] (Design and Analysis of Algorithms) — prereq: 6.1200[J] + 6.1210

Minimum path: 3–4 semesters depending on scheduling

DOWNSTREAM COURSES REQUIRING 6.1220[J]:

- 6.1400[J] Computability and Complexity Theory (also accepts 6.1220[J] OR 6.1400[J] — see note)
- 6.5310 Geometric Folding Algorithms: Prereq 6.1220[J]
- 6.5320 Geometric Computing: Prereq 6.1220[J]
- Various graduate (G) level 6.5xxx theory courses

CROSS-LISTING NOTE:

Students in Mathematics (Course 18) register as 18.400[J]. Both sections meet together. The credit and prerequisites are identical. If a downstream course lists 6.1220[J] as a prerequisite, completing 18.400[J] satisfies it equivalently.